Lecture 9 - Linearization and Derivative Rules

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10/08/2026

Post Exam Recap

1. Solutions to Exam 1 are up on Canvas, I'm happy to answer any questions about it if people want
2. Exams should be graded by tonight or tomorrow night, with curve details to follow after I release the scores
3. One important note so far: follow the instructions in the questions! Many people did not include reasoning for parts b and c of question 2; this will decrease your scores!

Quiz Time!

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Just kidding :) no quiz today

Linearization

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$$\mathscr{L}_{x=a}(x) = f(a) + f'(a)(x - a) \tag{2}$$

Using the Linearization to Approximate I

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Example 1: The *logistic population growth* model is a mathematical description of how the size of a population $P(t)$ grows over some time period t with some growth factor k and some carrying capacity N. It assumes that:

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Suppose that we have a population $P(t)$, where t is measured in days, which:

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What is our best estimate for the population size one year after we started measuring?

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And so:

$$P(1) \approx P(0) + P'(0)(1 - 0) = 1000 + 1000 \cdot 1 = 2000 \tag{9}$$

Using the Linearization to Approximate III

Is this an over-approximation ($\mathscr{L}_{t=0}(1) > P(1)$) or an under-approximation ($\mathscr{L}_{t=0}(1) < P(1)$)? What would we need to determine that?

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(Concave up => graph is above its tangent line)

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What is a value $a \approx 25.1$ where we know $\sqrt{a}$ and the derivative of the sqrt function at $x = a$?

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We should set $a = 25$ since $25 \approx 25.1$ AND $\sqrt{25} = 5$ is easy!

Then we get:

$$\sqrt{25.1} \approx \sqrt{25} + \frac{1}{2 \cdot \sqrt{25}}(25.1 - 25) \tag{11}$$

$$= 5 + \frac{1}{2 \cdot 5} \cdot (0.1) \tag{12}$$

$$= 5 + \frac{1}{10} \cdot 0.1 \tag{13}$$

$$= 5 + 0.01 = 5.01 \tag{14}$$

Using the Linearization to Approximate VI

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Well:

$$\frac{d^2}{dx^2}\Big|_{x=5}\sqrt{x} = \left(-\frac{1}{4\sqrt{x^3}}\right)\Big|_{x=5} \tag{16}$$

$$= -\frac{1}{4\sqrt{125}} < 0 \tag{17}$$

So this is an *over* approximation since we are concave down at $x = 5$.

Derivative Rules

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GROSS!!

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$$= \frac{d}{dx}\frac{1}{2}x^{-\frac{1}{2}} \tag{24}$$

$$= \frac{1}{2}\frac{d}{dx}x^{-\frac{1}{2}} \tag{25}$$

$$= \frac{1}{2}\cdot -\frac{1}{2}\cdot x^{-\frac{1}{2}-1} \tag{26}$$

$$= \frac{-1}{4\sqrt{x^3}} \tag{27}$$

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Atomic Rules:

- Power Rule: $\frac{d}{dx}x^n$
- Trigonometric Rules: $\frac{d}{dx}\sin(x)$ etc.
- Exponential/Logarithm Rules: $\frac{d}{dx}e^x$, $\frac{d}{dx}\ln(x)$

Combination Rules:

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Combination Rules:

- Sum/Difference Rule: $\frac{d}{dx}(f(x) \pm g(x))$
- Product Rule: $\frac{d}{dx}f(x)g(x)$
- Chain Rule: $\frac{d}{dx}g(f(x))$
- Constant Multiple Rule: $\frac{d}{dx}c \cdot f(x)$
- Quotient Rule: $\frac{d}{dx}\frac{f(x)}{g(x)}$
- Inverse Function Rule: $\frac{d}{dx}f^{-1}(x)$

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Perhaps the most straightforward derivative rule:

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$$(f(a) + f'(a)(x-a)) + (g(a)g'(a)(x-a)) = (f(a) + g(a)) + (f'(a) + g'(a))(x-a) \tag{29}$$

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Similarly:

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The derivative of a function multiplied by a constant is that constant multiplied by the derivative of the function:

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The factor must actually be a constant!

$$\frac{d}{dx}x^2 = 2x \neq x \cdot \frac{d}{dx}x = x \cdot 1 = x \tag{32}$$

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We can use this to compute derivatives of functions like:

$$f(x) = 5x^3 + 3x^2 + 2x + 5 \tag{33}$$

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Can we see a pattern?
Guess: $\frac{d}{dx}x^n = n \cdot x^{n-1}$.

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Why?

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So we know that $\frac{d}{dx}x^n = nx^{n-1}$!

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Power rule only applies to ***POWER FUNCTIONS*** i.e. $f(x) = x^c$ for some constant c.
It does NOT apply to *exponential functions* like $f(x) = c^x$! They may look similar, but their derivatives are VASTLY different!

Exponential Rule I

What is the derivative of an exponential function?

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What is the constant factor $\lim_{h\to 0}\frac{c^h-1}{h}$?

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Define a constant e so that $\lim_{h \to 0} \frac{e^h - 1}{h} = 1$ (i.e. $\frac{d}{dx} e^x = e^x$). Then:

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This gives us:

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The derivative of an exponential function, i.e. c^x for a constant c is:

$$\frac{d}{dx}c^x = \ln(c)c^x \tag{58}$$

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Exponential functions growing rapidly => it's derivative grows very large very quickly.
Very different behavior to that for power rule $\frac{d}{dx}x^c = cx^{c-1}$.

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